

Empowering UX with Web Components



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➤ Who are we?

Julie Xiong

Senior Front End Engineer

- Designer turned engineer
- Consult development teams on front end best practices
- UI design implementation
- Maintain design toolkit
- HTML & CSS queen
- Loathes IE11 with a fiery passion

Patrick O'Brien

Senior Software Engineer, Tech Lead

- Leads a team of developers for insurance-related software
- Enjoy getting hands dirty with new front end technologies
- IE11 is the spawn of Satan
- Helped architect web component implementation of Zywave's design toolkit

➤ Why are we here?

Designers and developers don't speak the same language

The platform gives us a lot of power for free. How can we best utilize this functionality?

Enforcing consistency of brand across disparate tech stacks is challenging

How can we build one design library to rule them all?

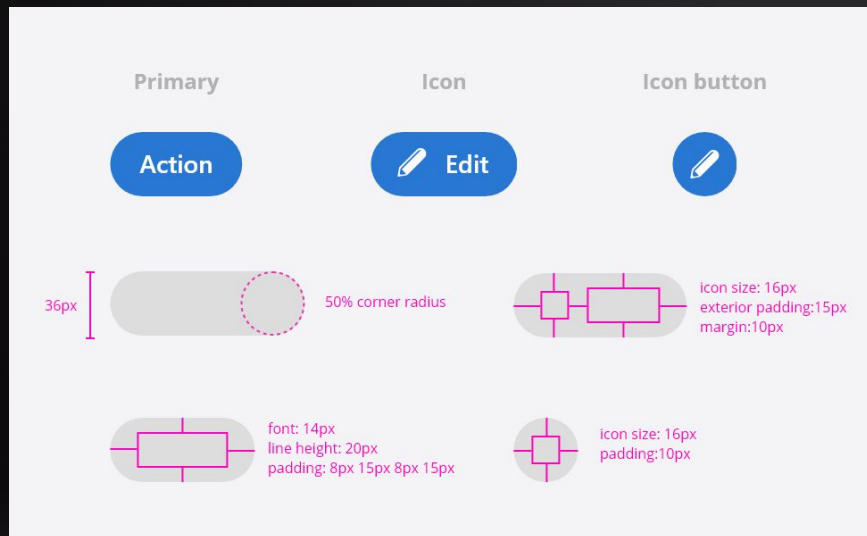
Maintaining a design library can be expensive

Who's going to maintain that monolithic CSS library?

➤ Zywave User Interface (ZUI)

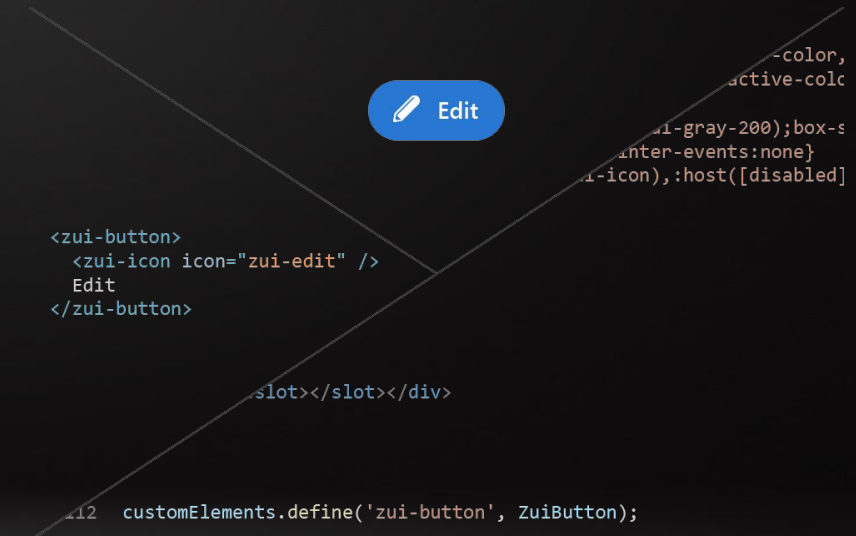
Design system

Defines design standards



Toolkit

Enforces design standards via Web Components



➤ Pain points of maintaining a design library

- Development bottleneck
- Design debt accrues whenever there are changes to design standards
- Painful to port toolkit for a new framework
- Waterfall method is bad
- Little to no developer involvement until it was too late

Design and development:

Speaking the same language

Designers

Quality of experience

Visuals

Usability testing

Beautiful user interfaces

Analytics

Brand values

User experience

Developers

Quantity of available features

Code quality

Efficiency

Web standards

Fast

Technical constraints

Functional

A painting of two muscular men wrestling. The man on the left is wearing a white shirt and a dark vest, while the man on the right is wearing a red shirt. They are both flexing their muscles, with their hands clasped together in the center. The background is a dark, moody landscape with a hint of a horizon. The text 'USER NEEDS' is written in white, bold, sans-serif font across the center. The text 'DESIGNERS' is written in white, bold, sans-serif font, slanted upwards from left to right, over the man in the white shirt. The text 'DEVELOPERS' is written in white, bold, sans-serif font, slanted upwards from left to right, over the man in the red shirt.

USER NEEDS

DESIGNERS

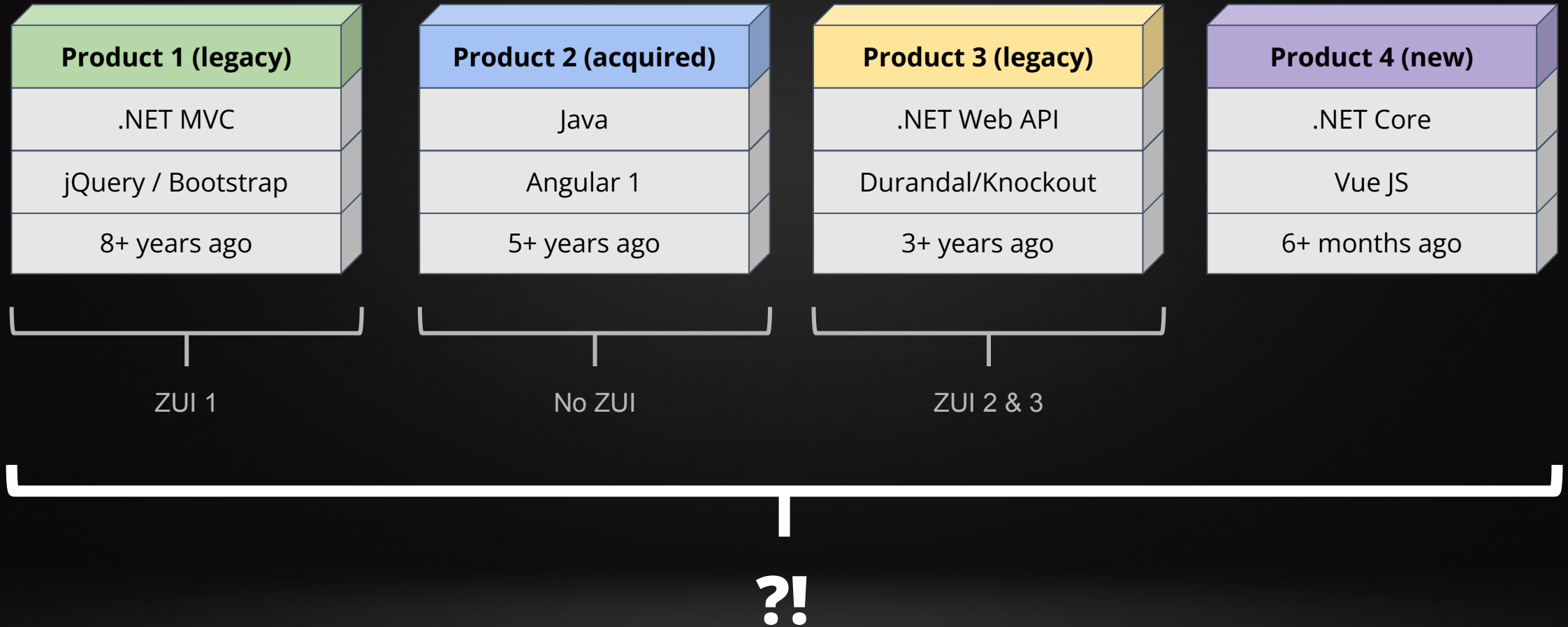
DEVELOPERS

➤ Collaborate with the user in mind

- Both take ownership of the design library
- Designers learn more about web standards
- Developers learn more about UX and design best practices
- Understanding each other's perspectives allows for better communication and collaboration
- Both parties will be more invested in building a kickass product

Building a scalable design system >

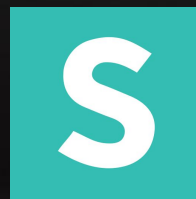
➤ Problem: All of the tech stacks, none of the resources



➤ Solution 1: CSS libraries

Pros

- Great browser support
- Common skillset
- Well documented
- Supported by every tech stack that runs in the browser



Cons

- Hard to build maintainable code base
- Lack of modular system creates a monolithic structure
- Easy for developers to mess up
- Can be difficult to use if complex components are created
- “Behaviors” immediately introduce JS extensions

➤ Solution 2: Framework-specific components

Pros

- Easy to build more complex components
- Design library and product framework are likely to be the same stack (easier to understand for all)

Cons

- Products are now handcuffed to the selected framework
- Upgrading/replacing framework might require a full rewrite
- Expensive to maintain if need to support multiple frameworks

➤ Solution 3: Web Components

Pros

- Works for any* tech stack
 - <https://custom-elements-everywhere.com/>
- Smaller JS footprint than a full framework
- Shadow DOM gives designers more control of the UI
- Eliminates code collisions
- Browser support:
 - Chrome, Firefox, Safari, Opera
 - Chromium Edge? 🙏

Cons

- Newer technology, less knowledge
- Vanilla web components are a bit difficult to write/understand
- No support for IE11 & Edge
 - Polyfills exist



CSS libraries



Framework-specific
components



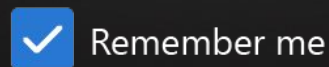
Web components

➤ What are Web Components?

- Browser supported “component model” popularized by frameworks like React
- Allows you to create reusable custom elements



```
<zui-button>  
  <zui-icon icon="zui-edit" />  
  Edit  
</zui-button>
```



```
<zui-checkbox checked>  
  Remember me  
</zui-checkbox>
```



```
<zui-search></zui-search>
```

➤ What makes a Web Component?

1. Custom elements
2. HTML templates (`<template>`)
3. Shadow DOM
4. ES modules
5. CSS custom properties
aka CSS variables

```
<?xml?><html>...</zui-checkbox>
<#shadow-root (open)>
  <head><div class="checkbox-container">
    <style>class="checkbox">
      <template>class="checkmark-container">
        --my-color: var(--checkbox-color);
        <div>checkbox-color: var(--my-color);
      } <div class="animate"></div>
    </style>
    <script type="module" src="main.js"></script>
  </head><div class="icon">
    <body> <slot name="icon"></slot>
    <div><zui-checkbox>...</zui-checkbox>
  </body><div class="label-text">
</html> <slot>...</slot>
  </div>
</div>
</section>
<style>...</style>
  "Unsubscribe from surveys"
</zui-checkbox>
```

Make your design system more resilient
with Web Components

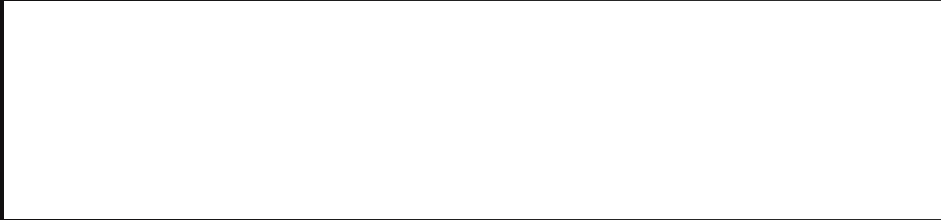


➤ Problem: design changes cause developer friction

- Design is subjective and changes with the times
- An application's HTML structure greatly depends on the layout/design
- Bugs are a reality; need to be able to fix them without causing issues

➤ Less developer friction with Web Components

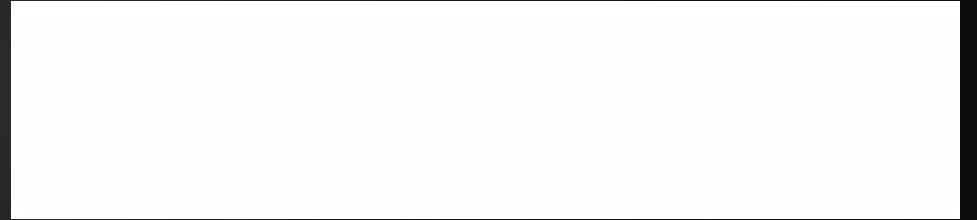
Old ZUI Well



```
<zui-well type="info">...</zui-well>
```

1. Container grow animation
2. Text fade in animation
3. Icon pop
4. Blue background

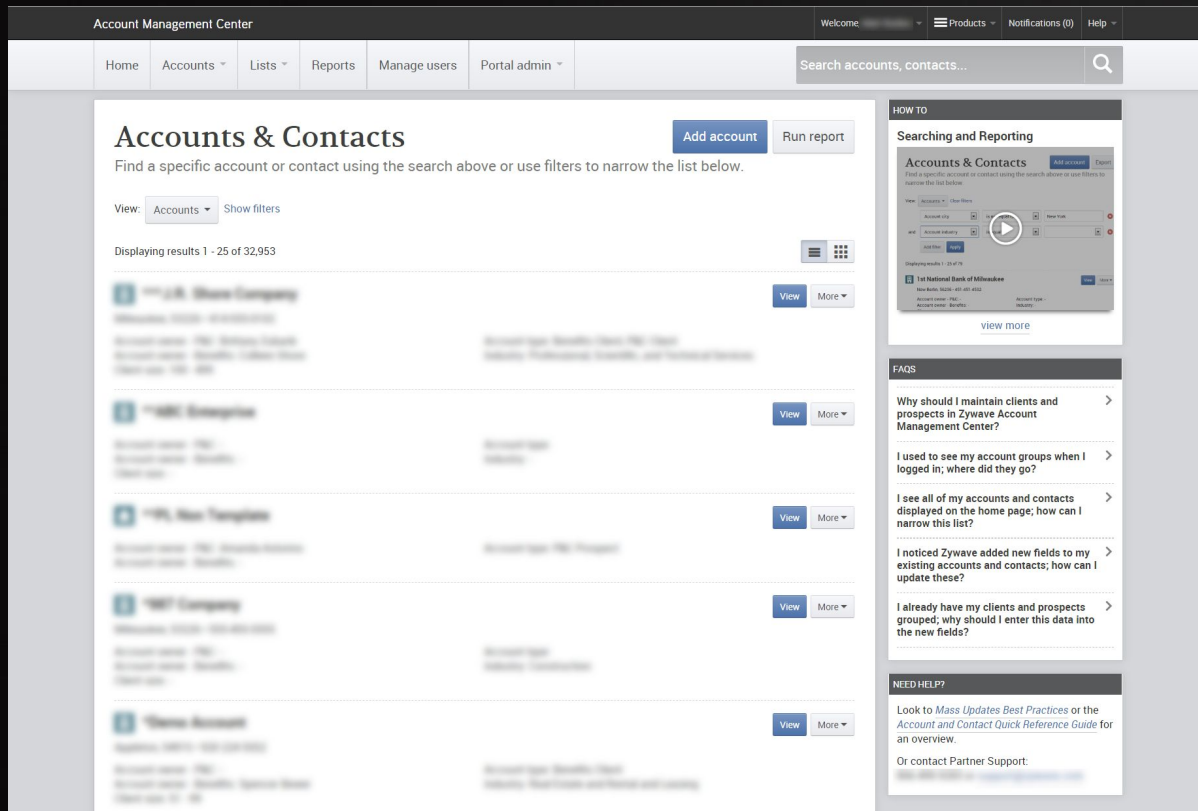
New ZUI Well



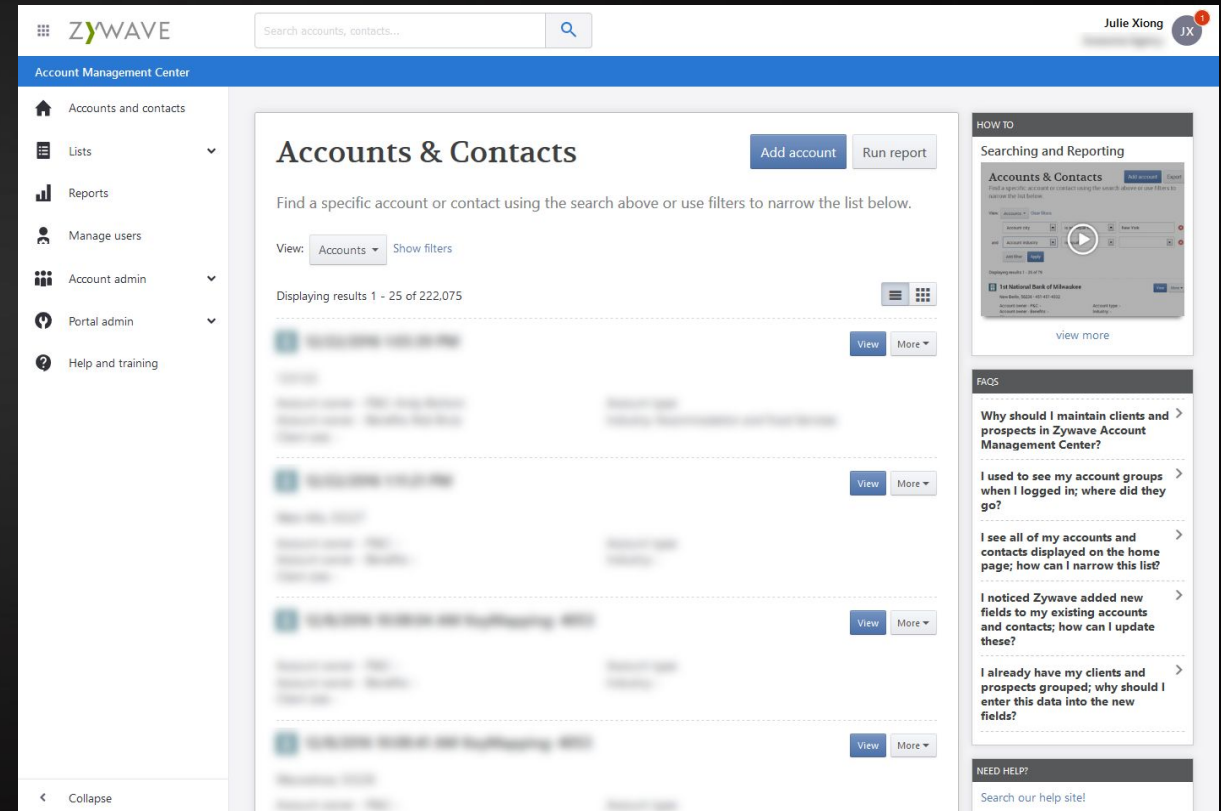
```
<zui-well type="info" dismissible>...</zui-well>
```

1. Container grow animation
2. Text fade in animation
3. Icon pop
4. Bar grow animation (new)
5. Dismissible feature (new)
6. Banner feature (new)
7. Static feature (new)
8. White background (new)

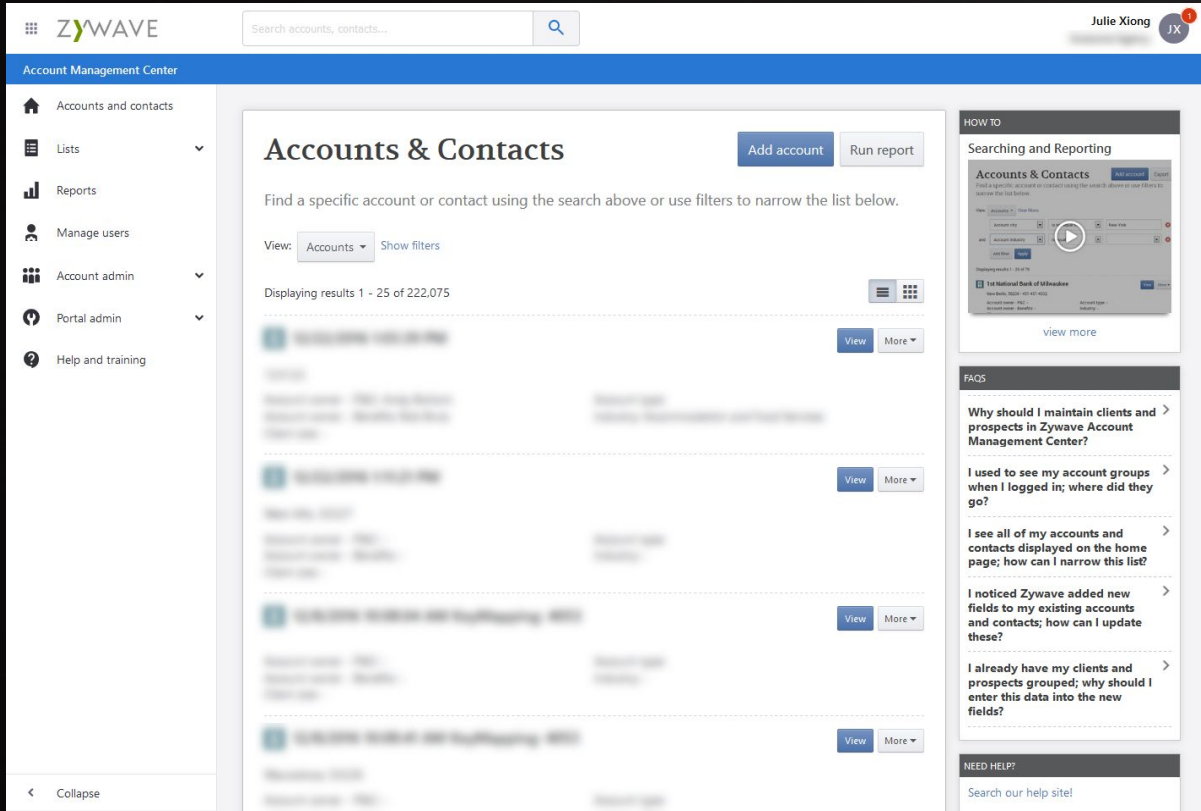
Web Components in practice >



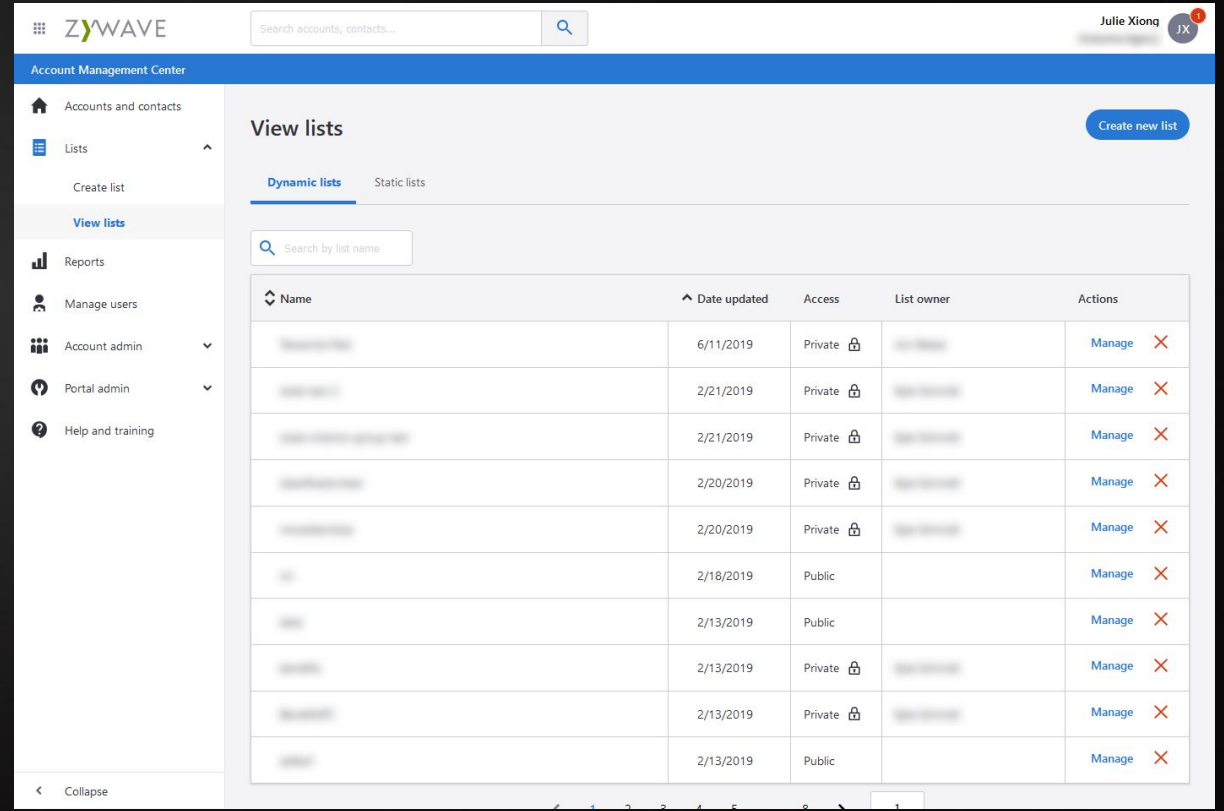
2012



2019



10 years old



10 months old

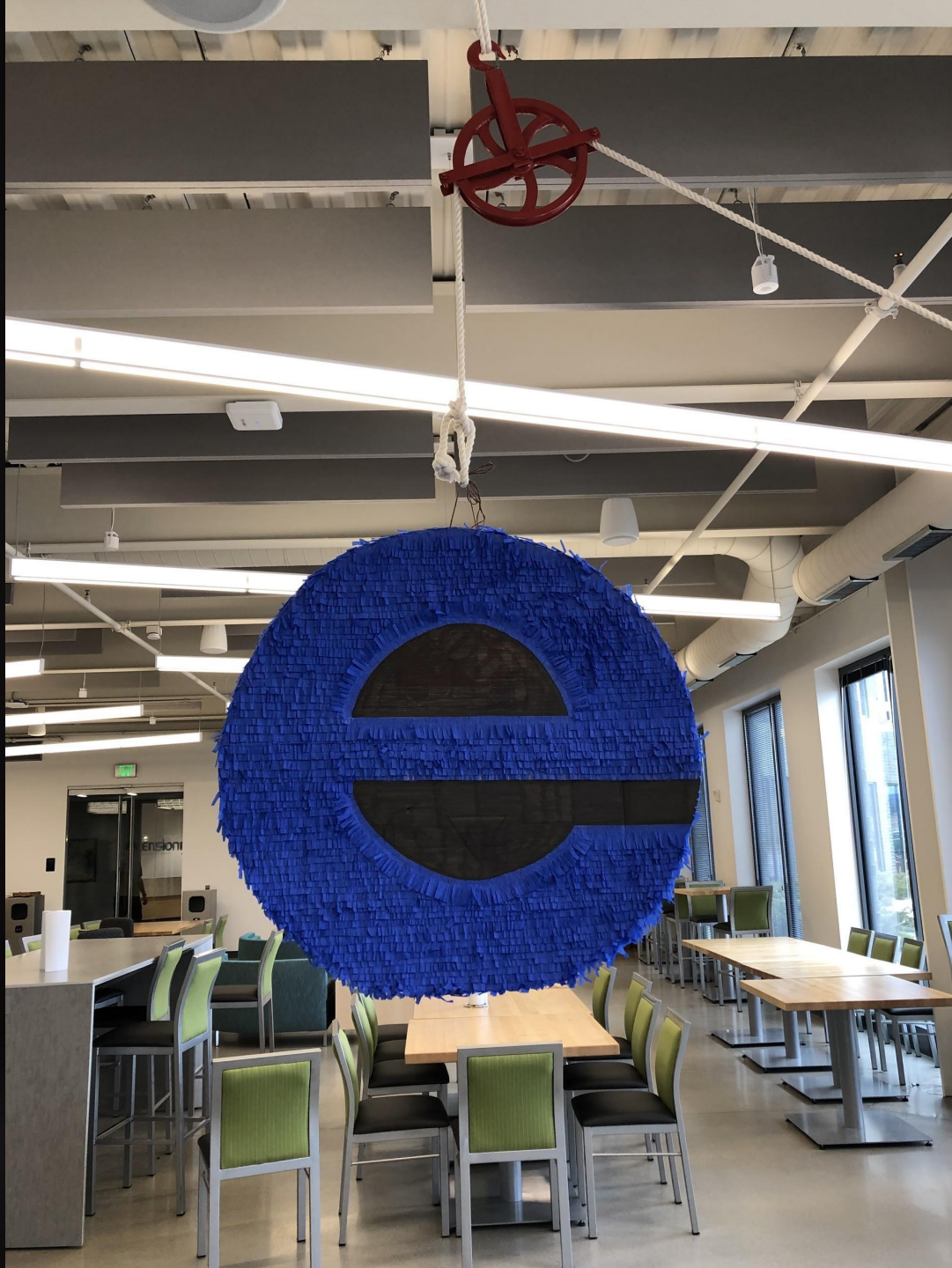
➤ Demos

- Vanilla web component:
<https://stackblitz.com/edit/ccc-vanilla-wc>
- Before and after web components:
<https://stackblitz.com/edit/zui-checkbox-demo>

➤ It's not all sunshine and roses...

- Purposefully hard to override design
- Custom element APIs are very low level; lots of boilerplate to get up and running
- “Bleeding edge”; still lots of new features coming and lots to learn
- If you use React, you're gonna have a bad time
- Internet Explorer is still a thing 😭





Is Web Components for me? >

➤ Yes, if you...

- Want to decrease design debt
- Like reusable components
- Spend too much time updating UIs
- Are tired of rewriting your design toolkit for every framework
- Need to maintain a scalable design library
- Don't care about IE11 (there are polyfills if you do)
- Like to learn new things

➤ Resources to get you started

Web Components

- [WebComponents.org](https://webcomponents.org)
- Getting started:
https://developer.mozilla.org/en-US/docs/Web/Web_Components
- Shadow DOM:
https://developer.mozilla.org/en-US/docs/Web/Web_Components/Using_shadow_DOM
- Custom CSS Properties:
https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties

lit-html and LitElement

- lit-html:
<https://lit-html.polymer-project.org/>
- LitElement:
<https://lit-element.polymer-project.org/>
- Polymer Slack:
<https://polymer.slack.com>

Slides: <http://bit.ly/ccc-wc>

➤ Pro tip: Treat your web components like self-contained APIs

- Attributes, events, public properties, and CSS should be kept as stable as possible
 - Being able to update the Shadow DOM, or a particular shade of red without consumers knowing is powerful
- Assume you know nothing about where the component will be used, and build defensively



Thank you!

Slides

<http://bit.ly/ccc-wc>

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